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Educational Design in the Adoption of ICT for Sustainable Digital Learning in Social and Business Sciences: A Structural Equation Model

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Abstract: Currently, teaching and learning are significantly influenced by Information and Communication Technologies (ICT), especially in a constantly evolving technological environment. The objective of this study was to develop a structural model to understand how university students use ICT for sustainable digital learning in the areas of social and business sciences. The study investigates the main factors influencing the acceptance of sustainable digital learning based on student satisfaction with the actual use of ICT. This research employs Structural Equation Modeling (SEM) with SEM-AMOS and utilizes a questionnaire based on the conceptual framework of the Technology Acceptance Model (TAM), its variants, and Social Cognitive Theory (SCT) as the primary method of data collection, from which 728 valid responses were obtained. Student responses were classified into several constructs and evaluated to determine their satisfaction with ICT for sustainable digital learning. The findings revealed a significant connection between computer self-efficacy, computer anxiety, and perceived enjoyment towards perceived usefulness and perceived ease of use, which in turn positively influenced student satisfaction with the use of ICT. This model proved effective in explaining student satisfaction and highlights the importance of deeply and strategically integrating ICT into educational design to promote digital learning as sustainability.

Keywords: ICT; TAM; sustainable digital learning; sustainability; student satisfaction; educational design



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1. Introduction

Globally, Information and Communication Technologies (ICT) have revolutionized higher education, facilitating access to knowledge and transforming teaching and learning methodologies [1]. With the advancement of the Internet and the World Wide Web, opportunities for educational access have greatly expanded, allowing students worldwide to leverage a wealth of educational resources from virtually anywhere [2,3]. These tools have become indispensable for educational activities and the design of didactic strategies [4]. However, for the effective incorporation of these technologies in the educational context, the digital capacities and perceptions of teachers must be aligned with this objective [5]. Likewise, the ICT infrastructure (software, hardware, networks, and media) employed throughout the learning and information exchange process must be available to achieve sustainable and enhanced learning outcomes [6].

The United Nations Educational, Scientific and Cultural Organization (UNESCO) highlights that ICT offers significant opportunities to enrich educational processes and promote lifelong sustainable learning [7]. In Ecuador, higher education has progressively integrated ICT in line with global efforts to improve educational quality and accessibility [8]. Although the availability of adequate ICT infrastructure remains a challenge, this issue is compounded by the lack of clear projection of its benefits in fostering sustainable education.

Factors such as policies, collaborations, faculty perceptions, and the adoption of flexible educational models are key, especially in developing countries, where investment does not always result in proportional improvements [9,10]. However, a significant part of the problem lies in the disparity between students' use of their personal technological resources and their understanding of university guidelines on the use of ICT for sustainable learning. Therefore, it is crucial to analyze how digital education can increase the effective use of ICT among students [10,11].

Research highlights that Information and Communication Technologies (ICT) play an important role in motivating students to learn by enhancing knowledge acquisition, retention, and comprehension, which inherently promotes the sustainability of higher learning [12]. ICT activates cognitive resources and makes learning more engaging, promoting a shift from teacher-centered to student-centered learning [13]. However, most studies do not delve into how these technologies influence less favorable contexts, such as in developing countries, where assessing teaching and learning effectiveness is crucial. This assessment not only determines the extent of ICT adoption by students but also affects their satisfaction with learning, a key aspect of university education sustainability [14]. Although satisfaction is widely recognized as an indicator of teaching quality [15], the literature on the subject remains limited. Additionally, the impact of ICT on student satisfaction may be influenced by several aspects of the educational environment, such as the mode of study (online or face-to-face), academic level (undergraduate or graduate), prior experience, frequency of use, and even demographic factors like gender or age [11,16–18].

Research shows that students generally have positive perceptions of Information and Communication Technologies (ICT) in the educational sphere. In the university context, ICT tools are perceived as particularly useful for learning disciplines such as mathematics, where factors such as user satisfaction and subjective norms play a crucial role in the adoption of these technologies [11]. High-performing students recognize the value of ICT in supporting their learning strategies, emphasizing its ability to enhance comprehension and educational effectiveness, thus contributing to more sustainable and enduring learning outcomes [19,20]. At the primary education level, both teachers and students report acceptable technological competence, although teachers indicate having more training and more intensive use of ICT compared to their students [21]. In secondary education, ICT is highly valued in science classes, where students feel these tools enrich their learning, facilitate understanding of complex concepts, and make classes more engaging.

In the context of higher education in Ecuador, the use of ICT has enabled educational institutions to offer more enriching learning experiences. However, recent studies suggest that despite ICT facilitating task completion and interaction with instructors, challenges persist regarding effectiveness across different environments, such as online, face-to-face, and at various academic levels [16,22,23]. Students appreciate the flexibility and support ICT offers but also acknowledge limitations, such as the digital divide and the need for more advanced technological training [24].

Some of the most emblematic theories explaining user acceptance of technology include the Theory of Reasoned Action (TRA), a widely used model for understanding and predicting human behavior. TRA postulates that behavioral intention is the best predictor of actual behavior, directly influenced by attitude toward the behavior and subjective norms [25,26]. Building on this foundation, the Technology Acceptance Model (TAM) was developed to explain and predict user acceptance of information technology, focusing on key constructs such as perceived usefulness, perceived ease of use, and behavioral intention to use [27,28]. On the other hand, the Unified Theory of Acceptance and Use of Technology (UTAUT) integrates elements from previous models, including TAM and TRA, proposing that performance expectancy, effort expectancy, and social influence predict behavioral intention while facilitating conditions and behavioral intention predict actual technology use [29,30].

The primary objective of this study is to analyze and understand how university students' perceptions of the effectiveness of Information and Communication Technologies

(ICT) influence their satisfaction with the use of these technologies in higher education. Specifically, this study seeks to measure and test the relationships between Computer Self-Efficacy (CSE), Computer Anxiety (CA), Perceived Enjoyment (PE), Perceived Usefulness (PU), Perceived Ease of Use (PEU), and Student Satisfaction (SS). To achieve this objective, a Structural Equation Modeling (SEM) approach is applied, a statistical technique that allows the simultaneous evaluation of multiple relationships between unobservable latent variables [31,32]. This approach is effective for exploring how the mentioned factors affect students' perceived satisfaction with using ICT in their learning processes. By focusing on how perceived satisfaction varies among students based on their perceptions and other underlying factors, this study aims to offer new approaches that can guide educational designs that effectively integrate ICT in the teaching of social and business sciences.

2. Theoretical Framework and Hypotheses Development

2.1. Integrative Theoretical Model (TAM and SCT) and Educational Design

The Technology Acceptance Model (TAM) and the Unified Theory of Acceptance and Use of Technology (UTAUT) are both prominent frameworks used to understand users' acceptance of technology. Both models aim to predict behavioral intention and actual use of technology by examining key influencing factors. On the one hand, TAM, developed by Davis et al. [33], is a theoretical framework driven by two main factors: perceived usefulness (PU), which refers to the extent to which a person believes that using technology will enhance their performance in a specific task, and perceived ease of use (PEU), defined as the degree to which a person believes that using technology will be free of effort [27,34]. On the other hand, UTAUT integrates elements from eight previous models, including TAM, and identifies performance expectancy, effort expectancy, social influence, and facilitating conditions as central predictors of technology usage behavior [29,35].

While both models consider the impact of users' perceptions on technology adoption, UTAUT offers a more comprehensive approach by incorporating additional social and facilitating factors [36]. However, the decision to employ TAM instead of UTAUT in this study is based on factors that strengthen its theoretical foundation, such as TAM's simplicity and parsimony, its extensive validation across various domains, including education, and its robust framework for analyzing the constructs CSE, CA, PE, PU, PEU, and SS [37]. Moreover, while UTAUT may provide a broader explanation, some authors argue that it lacks considerations regarding the technology adoption process and may hinder further theoretical development in research on this phenomenon [29,38].

The effectiveness of an educational design that promotes the adoption of technological tools depends on various variables. Within an accessible framework and considering available resources, it is essential to understand why students and teachers choose to use or not use technological tools in the teaching-learning process. Thus, by knowing the perceived usefulness and ease of use, key factors can be established to design attractive educational technologies [9–11,28]. Social Cognitive Theory (SCT) provides a flexible framework to understand and promote positive behavioral changes in various contexts, from personal development to health interventions and mass communication, emphasizing the reciprocal interactions between personal, behavioral, and environmental factors [39]. In this framework, we propose analyzing how university students in social and business sciences adopt technological tools for sustainable digital learning based on the theoretical approaches of TAM (and current contributions) and SCT. The adoption of Information and Communication Technologies (ICTs) has revolutionized teaching and learning methods. Research has shown that TAM factors significantly predict ICT adoption in educational settings, as well as the acceptance of digital learning, often outperforming demographic variables [40–42]. However, although the model is popular among researchers, it should be applied with caution due to its limitations in directly measuring perceived success [40,42].

2.1.1. Computer Self-Efficacy (CSE)

Computer Self-Efficacy (CSE) is defined as an individual's belief in their ability to effectively manage computers and the Internet, directly influencing their technological competence, outcome expectations, and emotional reactions to these technologies [43]. CSE is a significant determinant of the level of engagement with information technologies. This construct aims to assess students' confidence in specific skills such as completing educational tasks, exploring learning opportunities, communicating via email, and searching for information online, all of which are essential for effective learning. Although research on the impact of CSE on Perceived Ease of Use (PEU) and Perceived Usefulness (PU) yields mixed results, there is evidence of its positive influence on these constructs [44–46]. Based on these premises, we propose the following hypotheses:

H1: *The higher the CSE regarding ICT, the greater the PEU.*

H2: *The higher the CSE regarding ICT, the greater the PU.*

2.1.2. Computer Anxiety (CA)

Computer Anxiety (CA) is a widespread phenomenon characterized by feelings of apprehension, tension, and unease when using or anticipating the use of computer tools, and it can significantly affect job performance and academic success [47,48]. This construct measures the level of anxiety or fear that students experience when using computers and related technologies. The items range from a general fear of using computers and concern about making irreparable mistakes to discomfort and intimidation when using these technologies. Computer anxiety (CA) is a relevant issue in higher education, especially in an increasingly technology-driven environment. This anxiety can negatively influence the learning process, as well as the perceived acceptance, usefulness, and ease of use of technologies [13,47,48]. Additionally, it may lead students to question the viability of computer-based learning systems [13,35]. Based on this, the following hypotheses are proposed:

H3: *The higher the CA regarding ICT, the lower the PEU.*

H4: *The higher the CA regarding ICT, the lower the PU.*

2.1.3. Perceived Enjoyment (PE)

In the realm of ICT, perceived enjoyment refers to the pleasure a user experiences when using technology, irrespective of any performance outcomes, and is an influential factor in users' motivation to interact with technologies [6,49]. This construct aims to measure the pleasure and enjoyment students experience when using digital tools for learning, assessing the extent to which students find the technology-mediated learning process enjoyable and fun. Thus, perceived enjoyment captures an affective dimension of ICT use, which can significantly influence students' motivation and engagement with digital learning. Research on the acceptance of information technology by users has identified perceived usefulness and perceived ease of use as key determinants of perceived enjoyment, as it has a significant impact on satisfaction perception even when controlling for perceived usefulness [13,50]. Based on this, the following hypotheses are proposed:

H5: *The higher the PE regarding ICT, the greater the PEU.*

H6: *The higher the PE regarding ICT, the greater the PU.*

2.1.4. Perceived Ease of Use (PEU)

Perceived ease of use (PEU) refers to an individual's perception of the absence of effort required to use tools such as ICTs. This effort is assessed individually during the usage process and results in a perception that can change over time [27,34]. This construct explores how easy and accessible students find digital tools for learning and addresses an assessment of the clarity and logic in using digital tools, as well as the ease of using these tools at any time and place, with or without support. Research on perceived ease of use (PEU) has established various connections with other constructs in the field of ICT acceptance and utilization. In particular, PEU has been identified as a determining factor that positively influences perceived usefulness (PU) and student satisfaction (SS) when interacting with ICTs [23,34,50]. Based on this theoretical and empirical evidence, we present the following hypotheses:

H7: *The higher the PEU regarding ICT, the greater the PU.*

H8: *The higher the PEU regarding ICT, the greater the student satisfaction (SS).*

2.1.5. Perceived Usefulness (PU)

Perceived usefulness (PU) refers to the user's perception of whether a technology or system will be beneficial in achieving their goals and improving their efficiency or productivity in specific tasks [27]. In this study, perceived usefulness focuses on evaluating how students perceive that the use of digital tools can enhance their learning and academic development, primarily through the promotion of knowledge sharing and self-education. Studies have shown that PU has a strong positive effect on the degree of ICT use in teaching [23,50,51], although it does not always directly influence the adoption of these tools [51]. In this research, the following hypothesis is addressed:

H9: *The higher the PU regarding ICT, the greater the student satisfaction (SS).*

2.1.6. Students' Satisfaction (SS)

Perceived satisfaction (SS) is defined as the subjective evaluation that users make of their personal experiences following the acquisition and use of a service, product, or activity [52]. This concept reflects the extent to which users' expectations have been met or exceeded based on the perceived quality and effectiveness of the provided service or product [53]. The measure of satisfaction in this study focuses on gauging the perceived satisfaction of university students regarding the use of ICTs. The items address various aspects, such as the effectiveness of these tools for analyzing, sharing, and building knowledge, as well as overall student satisfaction with their use in an educational context. Research suggests that in the TAM model used in student satisfaction studies, satisfactory learning with the use of ICTs is primarily influenced by factors such as computer self-efficacy (CSE), perceived usefulness (PU), and perceived ease of use (PEU) [10]. Moreover, an extended TAM model that incorporates computer self-efficacy and subjective norms better explains students' intentions to use ICTs and their satisfaction [6]. Due to its explanatory power in representing a part of the acceptance of tools like ICTs, this construct was chosen as the response variable in this study [18,53], as shown in Figure 1.

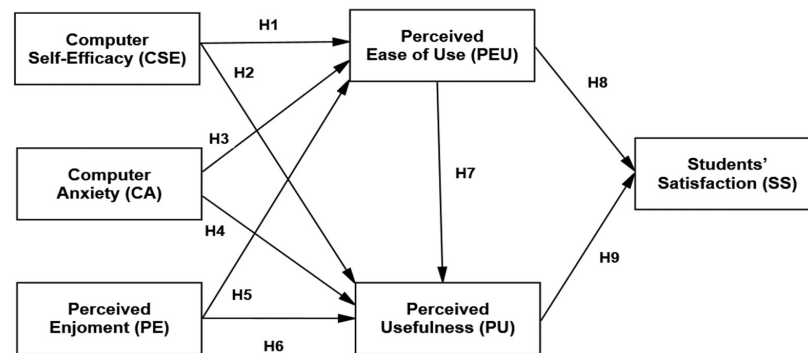


Figure 1. Research model.

3. Research Methodology

3.1. Research Design

The design of this research is empirical and employs a descriptive, cross-sectional approach to explore perceived satisfaction with the use of Information and Communication Technologies (ICT) among university students. This type of research was chosen for being the most reliable and for allowing immediate practical application due to its ability to obtain pertinent research results [54].

As such, this empirical study involved the participation of three faculties, Business Sciences, Education Sciences, and Philosophy, Letters, and Education Sciences, from three universities in Zone 5 of Ecuador. The selection criteria for the faculties were based on disciplinary specificity and the homogeneity of the sample [55], as well as the pedagogical need to thoroughly analyze perceptions of ICT use and the quality of technological experiences in these academic fields [56].

3.2. Data Collection

A total of 1252 students, randomly selected, with a margin of error of 2.72% and a confidence level of 95%, were included in the analysis. The study population comprised undergraduate and graduate students enrolled either online or in person in the faculties of Business Sciences, Education, or Social Sciences. Table 1 presents the composition of the sample, detailing the sociodemographic characteristics and the students' current situation regarding the use of ICTs.

Table 1. Sample distribution.

Category	Factor	Frequency	Proportion
Gender	Female	618	84.9%
	Male	110	15.1%
Age	18–21	148	18%
	22–25	138	19%
	26–29	130	19%
	30–33	135	18%
	More than 34	177	26%
Education level	Undergraduate	363	50.1%
	Graduate	365	49.9%
School	Business Sciences	250	34.3%
	Philosophy	241	33.1%
	Education Sciences	237	32.6%
Study Type	Online	465	63.8%
	In-person	263	36.2%
ICT Usage	Always	530	72.8%
	Sometimes	194	26.6%
	Never	4	0.5%
Duration of ICT Use	Less than 5 years	542	74.5%
	5–10 years	81	11.1%
	More than 10 years	105	14.4%

3.3. Procedure and Ethical Considerations

This study explores various aspects related to the perception of ICT use among university students. One of the main objectives was to gather a probabilistic, representative sample of students from various Ecuadorian universities. To achieve this, a survey was designed and administered using Google Forms, an effective tool for creating and distributing questionnaires. The surveys were electronically sent to students via their institutional emails, selected from three well-established email databases commonly used by academic directors for disseminating academic information. They were randomly distributed to a group of 1252 students from different regions of Ecuador. Subsequently, the Dillman method [57] was employed, sending electronic reminders to maximize the response rate. After the initial data collection, responses were carefully reviewed, and non-representative responses—such as those where participants provided the same rating across all items—were excluded, following the recommendations of Hair et al. [58] to ensure data quality and validity. This process resulted in 728 valid responses. In the initial phase of the survey, participants' previous experiences, regardless of their specific academic disciplines, were collected.

Various ethical considerations were taken into account following the guidelines of Winter and Gundur [59] to ensure the participants' well-being. Participants were informed about confidentiality, anonymity, and the protection of their privacy. Participation was entirely voluntary, and all participants provided informed consent prior to data collection. Additionally, to obtain more robust data, participants were provided with a clear explanation of the study's objectives and a basic conceptual description of ICT use in educational processes.

3.4. Measurement Instruments and Validity

This study explores various aspects of the perception of the use of Information and Communication Technologies (ICT) among university students. A structured survey was designed and administered to assess these perceptions, which includes five underlying constructs, each aimed at exploring a specific dimension of student perception towards ICT. This approach is based on previous studies such as those by Al-Rahmi et al. and Sayaf et al. [6,10], which identified five underlying constructs measuring Perceived Satisfaction (SS) adapted from [60], Computer Anxiety (CA) from [29], Computer Self-Efficacy (CS) from [43], Perceived Ease of Use (PEU) from [27] and Perceived Enjoyment (PE) from [61]. The survey consists of 19 items, each measured using a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree). The constructs, items and validity (internal consistency and reliability) are detailed in Table 2.

Table 2. Measurement model, factor loadings, reliability and internal consistency.

Factor	Item Code	Items Description	Loading	AVE	C.R.	C.A.
Computer Self-Efficacy	CSE1	I can learn to use computers for my digital learning process.	0.691	0.590	0.878	0.875
	CSE2	I can effectively complete my digital learning when I use computers.	0.763			
	CSE3	I can expand my learning opportunities by using computers and the Internet.	0.828			
	CSE4	I can use email (e.g., Hotmail, Outlook, Yahoo, Gmail) to learn to communicate.	0.733			
	CSE5	I can use computers and the Internet to search for information and resources.	0.818			
Computer Anxiety	CA1	I am afraid of using computers.	0.705	0.512	0.807	0.805
	CA2	I hesitate to use computers for fear of making mistakes that I cannot correct	0.777			
	CA3	Using computers and related technologies for learning makes me feel uncomfortable.	0.647			
	CA4	Computers and Internet technologies intimidate me slightly.	0.728			

Table 2. Cont.

Factor	Item Code	Items Description	Loading	AVE	C.R.	C.A.
Perceived Enjoyment	PE1	I find it pleasant to use digital tools for learning.	0.773	0.647	0.845	0.838
	PE2	The actual process of using digital tools for learning is enjoyable.	0.883			
	PE3	I have fun using digital tools to learn.	0.751			
Perceived usefulness	PU1	Using digital tools can improve my knowledge sharing.	0.839	0.652	0.848	0.843
	PU2	Using digital tools can enhance self-education.	0.840			
	PU3	Using digital tools can increase the effectiveness of my learning.	0.739			
Perceived ease of use	PEU1	I find digital tools easy to use at any time.	0.771	0.501	0.798	0.790
	PEU2	The use of any digital tool is clear and logical.	0.784			
	PEU3	I consider digital tools easy to use from anywhere.	0.706			
	PEU4	I can use any digital tool without problems if I have support.	0.544			
Students' Satisfaction	SS1	Digital learning tools are effective for knowledge analysis.	0.764	0.670	0.910	0.909
	SS2	I am pleased to use digital tools for learning.	0.786			
	SS3	Digital learning tools are effective for gathering knowledge.	0.846			
	SS4	Digital learning tools are effective for knowledge sharing.	0.845			
	SS5	Digital learning tools are effective for knowledge construction.	0.847			

Note. AVE, average variance extracted; C.R., Composite Reliability; C.A. Cronbach's alpha.

In this regard, Cronbach's alpha (α) values and composite reliability (CR) were calculated to verify the internal consistency of the constructs. Cronbach's alpha (recommended values between 0.7 and 0.8) measures the extent to which a set of items reflects a single unidimensional latent construct, while CR (recommended values > 0.7) indicates the reliability of the indicators measuring a specific factor. Although both Cronbach's alpha and CR indicate internal consistency, CR is preferred over Cronbach's alpha in SEM analysis at the construct level [58,62]. In this research, Cronbach's alpha values ranged from 0.79 to 0.90, and CR values ranged from 0.798 to 0.910 for all constructs, confirming that they exhibit acceptable internal consistency.

3.5. Model Fit Measures

The development and validation of the instruments and constructs used in this study followed a rigorous process. The content was validated through an exhaustive literature review to verify and select the most widely accepted items for the survey. The reliability and validity of each construct were tested through a series of analyses. The theoretical model chosen was specified and estimated using a Covariance-Based Structural Equation Model (CB-SEM) approach. The reliability and validity of the constructs were assessed through a measurement model utilizing a Confirmatory Factor Analysis (CFA) with IBM SPSS AMOS, version 23. Finally, the model fit was assessed through eight indices measuring discrepancy, incremental fit, parsimony, and adjusted relations [63] (see Table 3).

Table 3. The model fit indices.

Measures of Fit	Indices	Value	Recommended Values
Discrepancy measurement	CMIN/DF	2.892	(<2)
	Root Mean Square Error of the Approximation (RMSEA)	0.051	(0–0.1)
	Comparative Fit Index (CFI)	0.953	[0.9–1]
Incremental adjustment measures	Normed Fit Index (NFI)	0.931	[0.9–1]
	Tucker-Lewis Index (TLI)	0.946	(0.9–1)
	Incremental Fit Index (IFI)	0.953	(0.9–1)
Parsimony-adjusted and related measures	Parsimony-Adjusted Measures (PCFI)	0.819	(0.5–1)
	Goodness-of-Fit Index (GFI)	0.925	(0.9–1)

3.6. Data Analysis

For the analysis of the model's properties, the covariance-based structural equation modeling (CB-SEM) approach was applied using AMOS software (v. 23). Parameters were estimated through the maximum likelihood (ML) method [64]. Since there was no need to calculate fit measures with incomplete data, neither a saturated nor an independent model required adjustment. A convergence criterion limited to 50 iterations was established, and unbiased covariances were provided as input data [64].

In this study, the CB-SEM method in AMOS was used to obtain results such as standardized estimates, squared multiple correlations, total, direct, and indirect effects, factor loadings, and covariances of estimates. This approach is recommended for research objectives that involve testing a theoretical model; however, some authors suggest a complementary rather than a divisive use between users [65].

As part of the methodological process, both the measurement and structural models were evaluated. The structural model allowed for the examination of relationships between latent variables, while the measurement model focused on the reliability and validity of the constructs. The evaluation of the reflective measurement model revealed the indicator loadings, internal consistency reliability—which includes Cronbach's alpha and composite reliability—convergent validity through average variance extracted, and discriminant validity using the heterotrait-monotrait ratio (HTMT) and the Fornell-Larcker criterion. Meanwhile, the structural model was assessed via path coefficients and *p*-values [66]. Finally, due to the inherent nature of the 5-point Likert scale, there was a lack of normality in the data, which could bias the estimates and regression weights. To address this issue, the ML method was applied along with the bootstrapping technique, using 2000 bootstrap samples to obtain 95% bias-corrected confidence intervals [67].

4. Results

The theoretical model of this research was based on the use of Covariance-Based Structural Equation Modeling (CB-SEM), a multivariate statistical technique that allows for the analysis and comparison of relationships between constructs that are not directly observable [31]. CB-SEM was selected for its versatility in examining dependencies between latent variables, reducing errors, and accurately analyzing constructs [68]. The data analyzed in AMOS (version 23) resulted in two models: a measurement model (see Figure 2), which evaluates the reliability of the constructs, validity, and model fit, and a structural model (see Figure 3), which addresses the hypotheses related to the use of ICT for sustainable digital learning in social and business sciences.

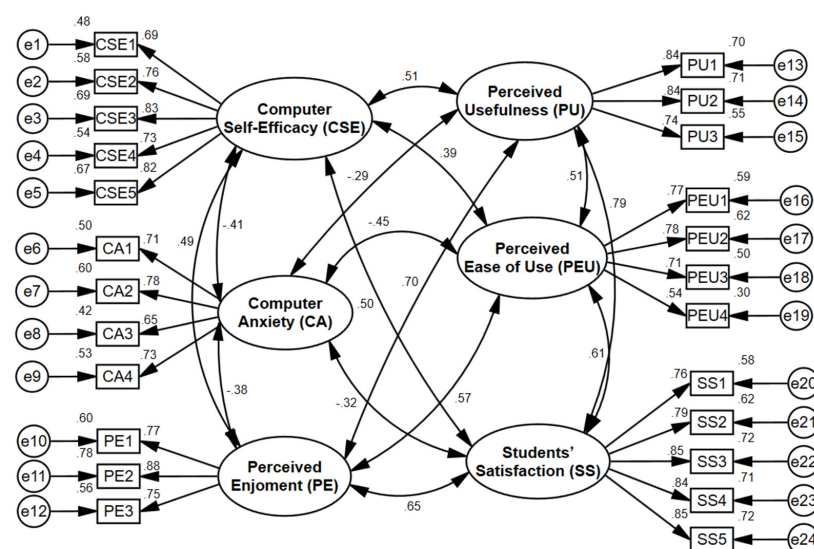


Figure 2. Measurement model.

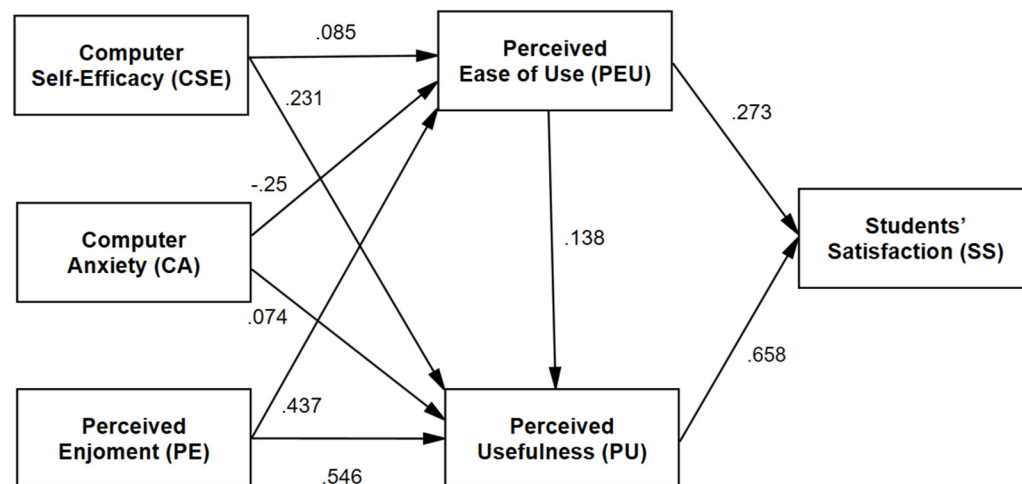


Figure 3. Path coefficient of Structural Model.

4.1. Reliability, Validity and Measurement Model Fit

The overall results confirm the validity of the measurement model as the basis of the structural model. This validity is corroborated by the reliability indicators of the items, where all reached values above the standard of 0.707, except for item PEU4 (0.54), which was kept for reasons of internal consistency since its elimination did not improve the model fit (See Figure 2). In addition, the Average Variance Extracted (AVE) was used to verify the proportion of variance explained by each construct compared to measurement error. The AVE values ranged from 0.501 to 0.67, exceeding the threshold of 0.5, indicating the presence of convergent validity (See Table 2).

The measurement model resulting from the Confirmatory Factor Analysis (CFA) shows that all constructs related to students' perceptions of ICT acceptance and use for sustainable learning present moderate to strong linear relationships. This is reflected in the model fit indicators, which yield acceptable values: CMIN/DF = 2.89, lower than the threshold of 5, and excellent values for TLI = 0.935, CFI = 0.988, and GFI = 0.935. Additionally, the assessment of the model's misfit to the observed data, measured by RMSEA, is satisfactory, with a value of 0.051, very close to the optimal threshold of 0.05.

Another key property of reliability and validity derived from the measurement model is discriminant validity, estimated using the Fornell and Larcker method [62]. This method involves comparing the square root of the Average Variance Extracted (AVE) with the correlations between constructs, where the diagonal value must be greater than the correlations between the constructs (see Table 4). In our study, the diagonal values are higher than those of the rows and columns corresponding to each measure, indicating that the relationship between a construct and its indicators is greater than the relationship of that construct with others. Discriminant validity between the constructs was also assessed using the HTMT (Heterotrait-Monotrait Ratio) criterion, as shown in Table 5. HTMT values are calculated by dividing the average correlation between different constructs (heterotrait) by the average correlation within the same construct (monotrait). The results obtained showed values below the recommended threshold of 0.85, indicating the presence of discriminant validity [69]. This means that each construct is distinct and reflects a unique phenomenon in comparison to the others.

Table 4. Discriminant validity is based on the Fornell and Lacker criterion.

	1.	2.	3.	4.	5.	6.
1. Computer Self-Efficacy	0.768	−0.414	0.489	0.508	0.395	0.5
2. Computer Anxiety	−0.414	0.716	−0.384	−0.293	−0.452	−0.318
3. Perceived Enjoyment	0.489	−0.384	0.804	0.699	0.57	0.649
4. Perceived usefulness	0.508	−0.293	0.699	0.807	0.51	0.785
5. Perceived ease of use	0.395	−0.452	0.57	0.51	0.708	0.605
6. Students' Satisfaction	0.5	−0.318	0.649	0.785	0.605	0.818

Note. The values in bold correspond to the square root of the AVE for each construct.

Table 5. Discriminant validity based on HTMT.

	1.	2.	3.	4.	5.
1. Computer Self-Efficacy					
2. Computer Anxiety	−0.433				
3. Perceived Enjoyment	0.489	−0.415			
4. Perceived usefulness	0.504	−0.305	0.716		
5. Perceived ease of use	0.415	−0.445	0.593	0.553	
6. Students' Satisfaction	0.500	−0.334	0.657	0.810	0.646

4.2. Structural Model

The complete structural model suggests that high student satisfaction (SS) with the use of ICTs for sustainable digital learning in higher education, particularly in social and business sciences, is derived from a combination of key factors. These include high computer self-efficacy (CSE) and high perceived enjoyment (PE), along with low computer anxiety (CA), all mediated by the perceived usefulness (PU) and perceived ease of use (PEU) of technological tools, as shown in Figure 3. These results highlight the importance of designing innovative educational approaches that integrate autonomous and digital learning, thus promoting the adoption of technologies that support more sustainable learning in complex environments [6,10].

Table 6 shows that CSE ($\beta = 0.085$, CR = 1.848, $p = 0.065$) has a positive, though not significant, impact on PEU, suggesting that Hypothesis 1 is not supported in this analysis. However, CSE ($\beta = 0.231$, CR = 5.604, $p < 0.001$) has a positive and significant impact on PU, supporting Hypothesis 2. On the other hand, CA ($\beta = -0.25$, CR = −5.395, $p < 0.001$) has a negative and significant impact on PEU, supporting Hypothesis 3. Although CA ($\beta = 0.074$, CR = 1.79, $p = 0.074$) has a positive impact on PU, this result is not statistically significant, thus not supporting Hypothesis 4.

Table 6. Estimates for the structural coefficients.

Hypotheses and Path				Standard Estimates	Standard Error	Critical Ratio	p -Value	Bootstrap Confidence (Sample = 2000)			
								Estimate	Lower	Upper	p -Value
H1:	PEU	←	CSE	0.085	0.042	1.848	0.065	0.085	−0.037	0.196	0.162
H2:	PU	←	CSE	0.231	0.033	5.604	***	0.231	0.127	0.337	0.001
H3:	PEU	←	CA	−0.25	0.044	−5.395	***	−0.25	−0.349	−0.133	0.001
H4:	PU	←	CA	0.074	0.034	1.79	0.074	0.074	−0.01	0.164	0.077
H5:	PEU	←	PE	0.437	0.052	8.796	***	0.437	0.318	0.566	0.001
H6:	PU	←	PE	0.546	0.047	10.413	***	0.546	0.397	0.683	0.001
H7:	PU	←	PEU	0.138	0.041	2.926	0.003	0.138	−0.001	0.269	0.037
H8:	SS	←	PEU	0.273	0.031	7.272	***	0.273	0.149	0.391	0.001
H9:	SS	←	PU	0.658	0.044	14.365	***	0.658	0.545	0.775	0.001

Note. ***, $p < 0.001$.

Regarding PE, the analysis shows that PE ($\beta = 0.437$, CR = 8.796, $p < 0.001$) has a positive and significant impact on PEU, supporting Hypothesis 5. Additionally, PE ($\beta = 0.546$,

CR = 10.413, $p < 0.001$) also has a positive and significant impact on PU, supporting Hypothesis 6. The relationship between PEU and PU is also positive and significant ($\beta = 0.138$, CR = 2.926, $p = 0.003$), supporting Hypothesis 7. With respect to student satisfaction (SS), PEU ($\beta = 0.273$, CR = 7.272, $p < 0.001$) has a positive and significant impact on SS, supporting Hypothesis 8. Similarly, PU ($\beta = 0.658$, CR = 14.365, $p < 0.001$) has a strong positive impact on SS, supporting Hypothesis 9.

Finally, the bootstrapping technique was used to calculate confidence intervals, which improved the precision of the estimates and supported the stability of the estimators, indicating that the observed significant relationships are not due to chance. For example, for H1, it is confirmed that the effect of CSE on PEU is not significant ($p > 0.05$). In the case of H2, the confidence interval (0.127, 0.337) reaffirms the stability of the positive influence of CSE on PU. Similarly, for H9, the confidence interval (0.545, 0.775) demonstrates that the influence of PU on SS is consistent and significant.

4.3. Evaluation and Identification of Factors

Descriptive statistics, such as measures of central tendency and dispersion (Frequency, Mean, and Standard Deviation), are essential for summarizing and presenting the most important information and findings about the population and phenomena studied, making survey data more understandable and interpretable [70]. Although often considered less complex than sophisticated statistical methods, descriptive statistics provide key information that can be of great value to researchers, especially in meta-analytic studies that seek to identify potential moderations in the relationships between variables [70,71].

Table 7 shows the descriptive values for the study population, where responses are classified on a 1-to-5 scale (1 = “Strongly disagree”, 2 = “Disagree”, 3 = “Neutral”, 4 = “Agree”, 5 = “Strongly agree”), with “f” representing the frequency of each response.

Table 7. Measurement of Constructs.

Scores		1	2	3	4	5	Mean	Standard Deviation
Factor	Code	f (%)	f (%)	f (%)	f (%)	f (%)		
Computer Self-Efficacy	CSE1	34 (4.7)	3 (0.4)	48 (6.6)	317 (43.5)	326 (44.8)	4.23	0.947
	CSE2	16 (2.2)	4 (0.5)	79 (10.9)	410 (56.3)	219 (30.1)	4.12	0.785
	CSE3	24 (3.3)	6 (0.8)	46 (6.3)	317 (43.6)	335 (46)	4.28	0.88
	CSE4	23 (3.2)	4 (0.5)	58 (8)	316 (43.4)	327 (44.9)	4.26	0.876
	CSE5	26 (3.6)	2 (0.3)	24 (3.3)	244 (33.5)	432 (59.3)	4.45	0.871
Computer Anxiety	CA1	232 (31.9)	302 (41.5)	150 (20.6)	40 (5.5)	4 (0.5)	2.01	0.892
	CA2	166 (22.8)	259 (35.6)	180 (24.7)	105 (14.4)	18 (2.5)	2.38	1.06
	CA3	226 (31.1)	346 (47.5)	95 (13)	47 (6.5)	14 (1.9)	2.01	0.935
	CA4	274 (37.6)	303 (41.6)	94 (12.9)	50 (6.9)	7 (1)	1.92	0.928
Perceived Enjoyment	PE1	18 (2.5)	11 (1.5)	38 (5.2)	285 (39.2)	376 (51.6)	4.36	0.85
	PE2	11 (1.5)	3 (0.4)	49 (6.7)	358 (49.2)	307 (42.2)	4.3	0.743
	PE3	14 (1.9)	6 (0.8)	74 (10.2)	305 (41.9)	329 (45.2)	4.28	0.827
Perceived usefulness	PU1	12 (1.6)	4 (0.5)	36 (4.9)	340 (46.7)	336 (46.2)	4.35	0.747
	PU2	11 (1.5)	9 (1.2)	37 (5.1)	298 (40.9)	373 (51.2)	4.39	0.772
	PU3	10 (1.4)	3 (0.4)	73 (10)	356 (48.9)	286 (39.3)	4.24	0.758
Perceived ease of use	PEU1	10 (1.4)	30 (4.1)	186 (25.5)	358 (49.2)	144 (19.8)	3.82	0.843
	PEU2	5 (0.7)	46 (6.3)	232 (31.9)	326 (44.8)	119 (16.3)	3.7	0.84
	PEU3	8 (1.1)	47 (6.5)	195 (26.8)	329 (45.2)	149 (20.5)	3.77	0.883
	PEU4	11 (1.5)	29 (4)	151 (20.7)	374 (51.4)	163 (22.4)	3.89	0.846
Students' Satisfaction	SS1	7 (1)	6 (0.8)	96 (13.2)	431 (59.2)	188 (25.8)	4.08	0.71
	SS2	7 (1)	8 (1.1)	80 (11)	423 (58.1)	210 (28.8)	4.13	0.716
	SS3	5 (0.7)	5 (0.7)	72 (9.9)	431 (59.2)	215 (29.5)	4.16	0.677
	SS4	8 (1.1)	6 (0.8)	65 (8.9)	421 (57.8)	228 (31.3)	4.17	0.71
	SS5	5 (0.7)	5 (0.7)	82 (11.2)	422 (58)	214 (29.4)	4.15	0.689

Our results highlight that in the case of Computer Self-Efficacy (CSE), around 88% of students feel capable of learning and using computers and the Internet for their digital learning process, demonstrating confidence in technological skills and promoting effective ICT adoption in their academic training. On the other hand, Computer Anxiety (CA) shows that more than 70% of participants disagree or strongly disagree with experiencing fear or discomfort when using computers, indicating low levels of technological anxiety. Regarding Perceived Enjoyment (PE), approximately 90% of students find using digital tools enjoyable and fun, which increases their motivation and engagement in digital learning, a crucial element for the design of effective educational strategies.

Table 7 also shows that more than 90% of students recognize the Perceived Usefulness (PU) of ICT, acknowledging these tools as enhancers of knowledge sharing, self-education, and learning effectiveness. Regarding Perceived Ease of Use (PEU), 70% of respondents agree that digital tools are easy to use, especially when support is available. Finally, in terms of Student Satisfaction (SS), more than 85% of students are satisfied with the digital tools for various aspects of learning, such as knowledge analysis, gathering, and sharing, implying that ICT meets students' expectations, encouraging their continued use.

5. Discussion and Theoretical Implications

In the specific context of social and business sciences, our findings reinforce the need to incorporate technological tools that not only facilitate teaching but also foster knowledge and self-management and adapt to rapid technological changes and the socio-economic environment. However, we argue that creating sustainable educational ecosystems requires more than simply adopting ICT; it demands a deep and strategic integration that promotes critical thinking, digital collaboration, and the resolution of complex problems. These elements are essential for shaping professionals capable of addressing contemporary challenges, safeguarding the well-being of the teaching-learning process, and maintaining higher education institutions' connection with society through the effective use of ICT [8,72].

This study explores the relationship between student satisfaction with ICT use in social and business sciences and the effect of their levels of Computer Self-Efficacy (CSE), Computer Anxiety (CA), and Perceived Enjoyment (PE) when mediated by Perceived Usefulness (PU) and Perceived Ease of Use (PEU). The study focuses on an under-explored area, particularly in a Latin American context, where limitations and availability may pose challenges [8]. Our findings provide evidence of how higher education in these fields can benefit from ICT use to clarify an educational design that achieves constant adaptation for sustainable digital learning. However, we emphasize that the generalization of our results is more oriented towards a Latin American context due to gender inequality in our sample, a pattern observed in similar studies involving data collection through surveys [73,74]. Although there is evidence that no significant gender differences exist when explaining an SEM model [75], results from studies with a more homogeneous sample may identify more than one significant effect (for example, H1 and H4 in this study) [6,10].

The following outlines the main theoretical contributions of our research that support Technology Acceptance Models (TAM) and Social Cognitive Theory (SCT) while also offering new perspectives and advancing knowledge in the educational field and its design.

The first contribution of this study is to open a debate on the impact of teachers' and students' decisions in adopting ICT, as these are a determining factor in educational design that promotes sustainable learning. While [56,76] argue that the use of ICT in social sciences education is more related to Perceived Usefulness (PU) based on the amount of previous experience and not necessarily Computer Self-Efficacy (CSE) and that ICT system characteristics do not significantly influence Perceived Ease of Use (PEU) for more accessible digital learning, our findings present a different perspective. Our model suggests that it is crucial to improve factors such as Perceived Ease of Use (PEU) and Perceived Usefulness (PU) through Computer Self-Efficacy (CSE), thus enhancing students' digital competencies. Additionally, it is essential to implement educational strategies that reduce

reluctance or fear towards using technology by developing systems that are easy to use, ensuring that the tools are useful, simple, and intuitive and that their functionalities align with students' needs [50,51]. This implies that academic institutions must continually reconsider their educational designs to select the most appropriate approach to promote successful ICT adoption by students [14,21]. To achieve this, it is necessary to assess whether it is more effective to centralize responsibility on educators or foster a synergistic participation between teachers and students in this process [72].

The second theoretical contribution is to provide a perspective that allows for a better understanding of the independent and mediating roles of Perceived Usefulness (PU) and Perceived Ease of Use (PEU) as important antecedents in promoting student satisfaction in higher education in social and business sciences. Although studies [13,34,46,50] have reported on the importance of Perceived Usefulness (PU) and Perceived Ease of Use (PEU) and their significant impact on student satisfaction, the literature lacks specific details on the educational decisions and designs that best promote or sustain this satisfaction over time. It has been shown that ICT management in higher education with a sustainable focus has grown exponentially [77], changing the nature of teaching and learning environments. However, the integration of digital technologies in education for sustainable development requires a total transformation in approaches, methods, and, ultimately, in educational design, particularly in integrating practices that contribute to sustainability [77,78]. To achieve this goal, detailed steps must be followed, starting fundamentally by incorporating sustainable information concepts into social and business science programs and broadly using ICT to promote sustainability.

Based on this, we conclude and suggest that adopting ICT in social sciences education requires more than simply implementing technology. For students, it is essential that ICT be perceived as useful and easy to use to increase its adoption in e-learning systems [76]. Moreover, from the educators' perspective, it is crucial that theoretical ICT training is aligned with practical fieldwork so that they can effectively apply these technologies in their environments and contribute to sustainable academic development [77,78]. These ideas can be extrapolated to the context of our study, where it is important to strengthen ICT training in the teaching-learning field (teacher-student), ensuring that technology is perceived as an effective and practical tool for sustainable digital learning.

Finally, this study expands the existing literature on the Technology Acceptance Model (TAM) and Social Cognitive Theory (SCT) by demonstrating how Computer Self-Efficacy (CSE) and Perceived Enjoyment (PE) directly influence Perceived Usefulness (PU) and Perceived Ease of Use (PEU), which in turn affect Student Satisfaction (SS) within the context of social and business sciences. Furthermore, by focusing on a Latin American context, this study addresses a geographical gap in research on ICT adoption, offering unique perspectives on how cultural and socioeconomic factors can moderate the relationships proposed by TAM and SCT. These contributions not only validate and refine existing theoretical models but also provide a foundation for the development of more effective educational strategies that promote sustainable digital learning.

6. Limitations and Further Research

Despite the interest in employing robust statistical techniques, such as structural equation modeling, to evaluate study objectives, certain considerations are recognized. One of these is the representativeness of the sample, which, while including Ecuador's Zone 5 and encompassing three universities, could be expanded in future studies to further generalize the results to other regions or countries. Additionally, although the sample consisted of 728 valid responses from students' representative of each social and business sciences faculty, the high female participation (84.9%) offers a limited perspective. Thus, achieving a more balanced gender distribution in future research would be enriching, allowing for a deeper analysis. Nevertheless, we caution about the persistent gender differences in educational and professional preferences. Research reveals that women are overrepresented in social sciences, arts, humanities, and law, while men predominate in

natural sciences, mathematics, and engineering [73,79]. These trends are present in various countries and periods—especially in Latin America [74]. In this context, observable gaps exist in disciplines such as engineering, with a male majority, and in social sciences and medical services, with greater female representation [73,80]. These preferences may be influenced by sociocultural factors, gender stereotypes, and perceptions about job stability and employability [79].

Lastly, while the study focused on measuring perceptions and their relationship with other latent factors, incorporating additional data, such as the quality of ICT in universities (e.g., the number of computers per student, the speed and reliability of networks, and access to technical support), could complement the current findings. These measures would provide additional indicators to assess not only the level of satisfaction but also the quality of the educational, and technological environment.

7. Conclusions

This study validated the Technology Acceptance Model (TAM) in the context of higher education in Ecuador, providing data on students' perceptions and expectations regarding the use of Information and Communication Technologies (ICT) in their learning process. The theoretical and practical implications of the research were discussed, highlighting how ICT can promote more sustainable and effective learning in the social sciences and business fields. The findings emphasize the advantages of TAM, unveiling new insights into the acceptance and adoption of ICT by university students. It was found that ICT significantly contributes to enhancing the quality of learning activities, facilitating the acquisition, retention, and comprehension of knowledge. However, it was identified that, in the Ecuadorian and broader Latin American context, challenges related to aligning the digital perceptions and skills of teachers and students still persist, which may limit the successful implementation of ICT.

This study makes a significant contribution by demonstrating that TAM is sufficiently robust to explain how factors such as Computer Self-Efficacy (CSE), Computer Anxiety (CA), Perceived Enjoyment (PE), Perceived Usefulness (PU), and Perceived Ease of Use (PEU) affect Student Satisfaction (SS). Additionally, the importance of considering these factors to design educational strategies that effectively integrate ICT into the teaching of social sciences and business is underscored. The research model was built around nine hypotheses and focused on analyzing the relationships between the mentioned factors. The results confirmed most of the hypotheses proposed (except hypotheses H1 and H4), evidencing the positive influence of self-efficacy, perceived enjoyment, and perceived usefulness on student satisfaction. Nevertheless, mixed results were observed concerning computer anxiety and its impact on perceived usefulness, suggesting the need for further investigation in this area.

It is recommended that future research integrate cultural aspects into the model to enable cross-validation across different cultural contexts. This approach would allow for adapting and enhancing ICT integration strategies in higher education, promoting a more inclusive and effective learning experience globally.

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